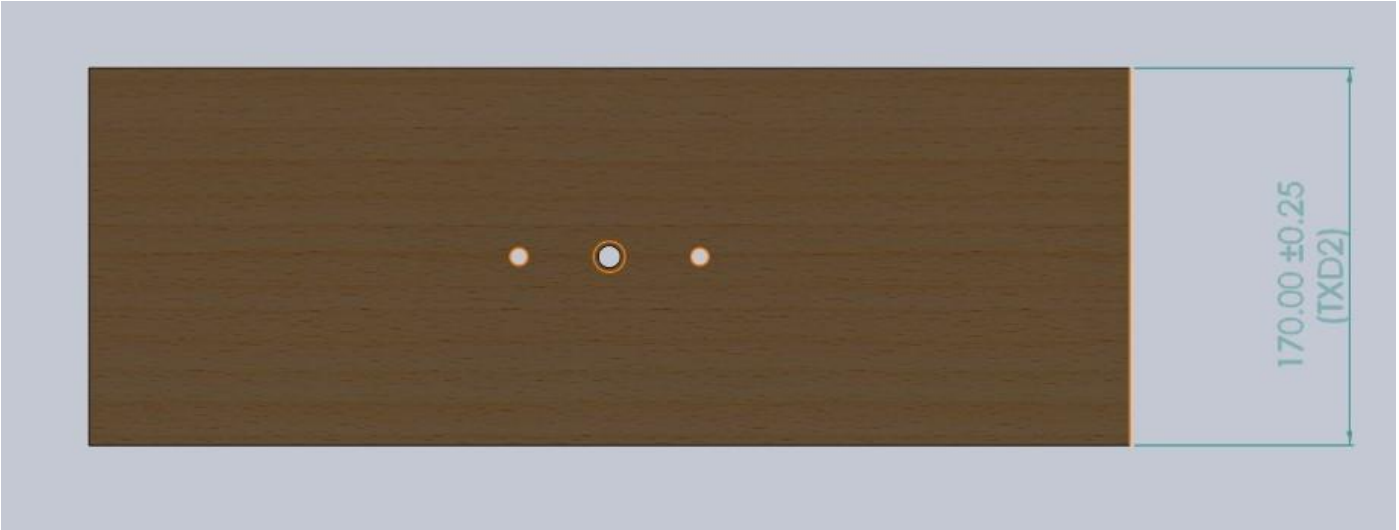
MANUFACTURING SCIENCE AND TOOLING

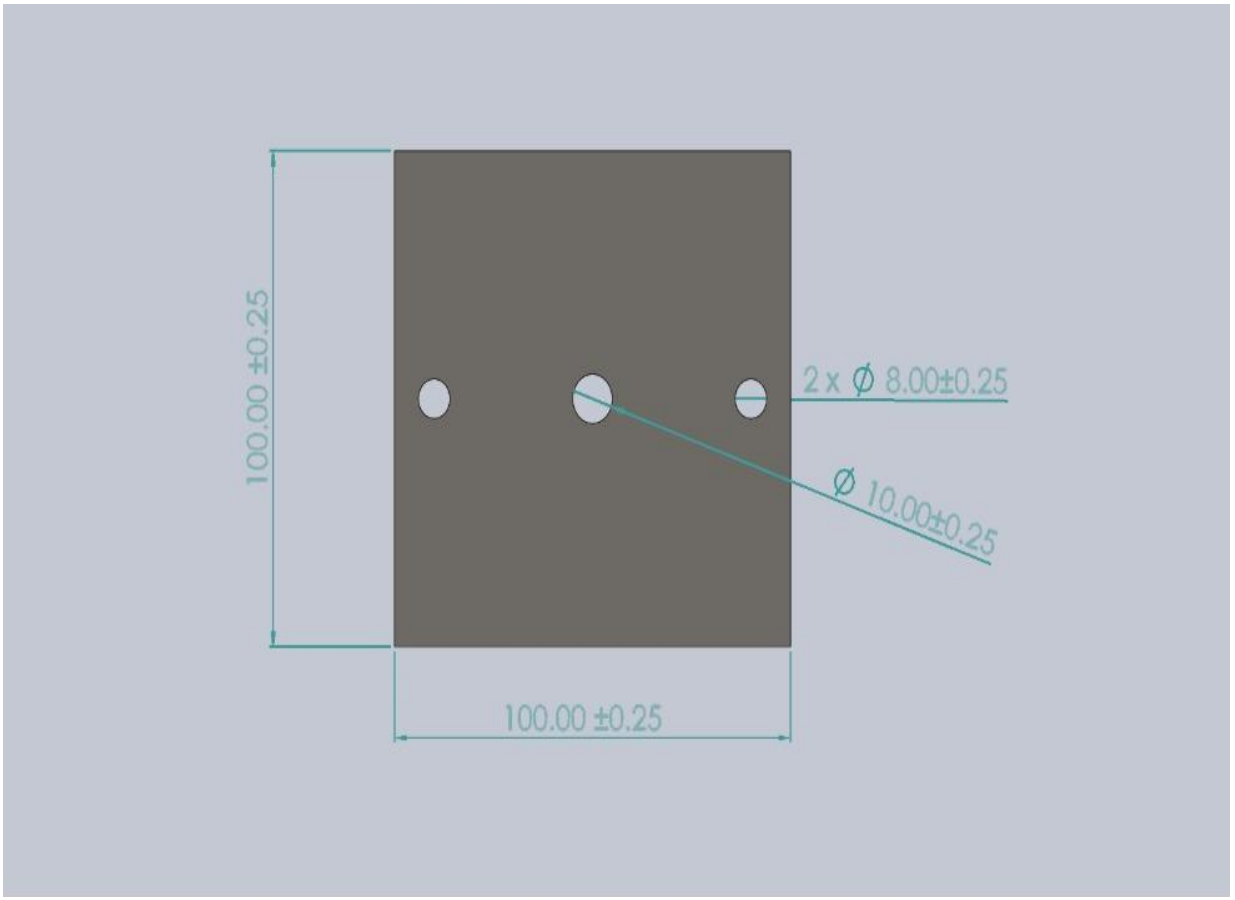
360 DEGREE ROTATING DRILLING MACHINE

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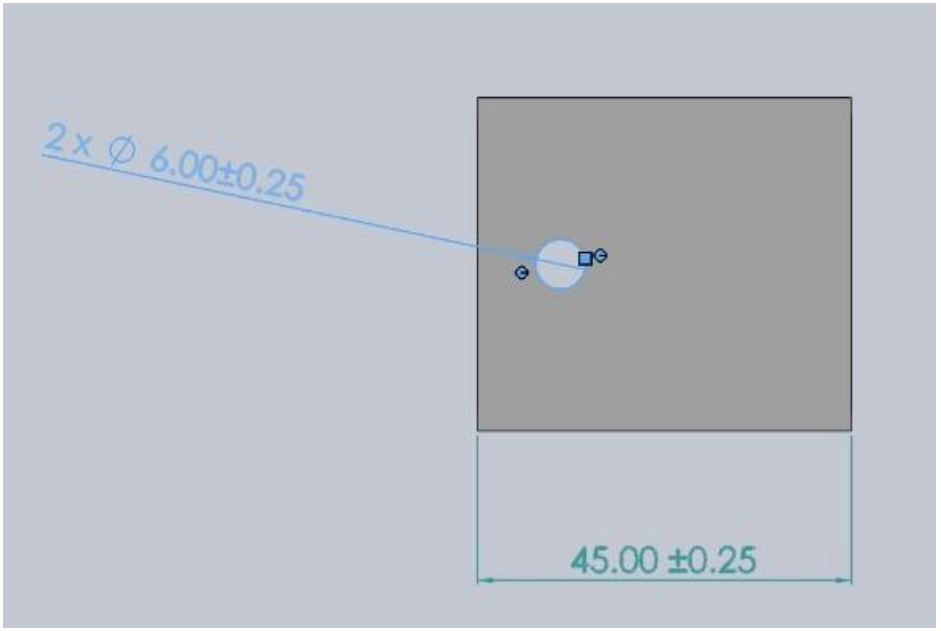
Sr. no.	Part	Operations	Dimensions (lxbxh) (In mms)	Machine used
1.	Base Plate	A. Taking wooden plate and cutting to required dimensions using saw B. Drilling 3 holes of required diameter to bolt metal base plate C. Using a chisel, to make the hole larger to fit the nut.	A. Plate:460x120x20 B. Holes:Dia 6 (2 holes) and Dia 10 (1 hole)	A. Hacksaw cutter B. Drilling Machine C. Chisel

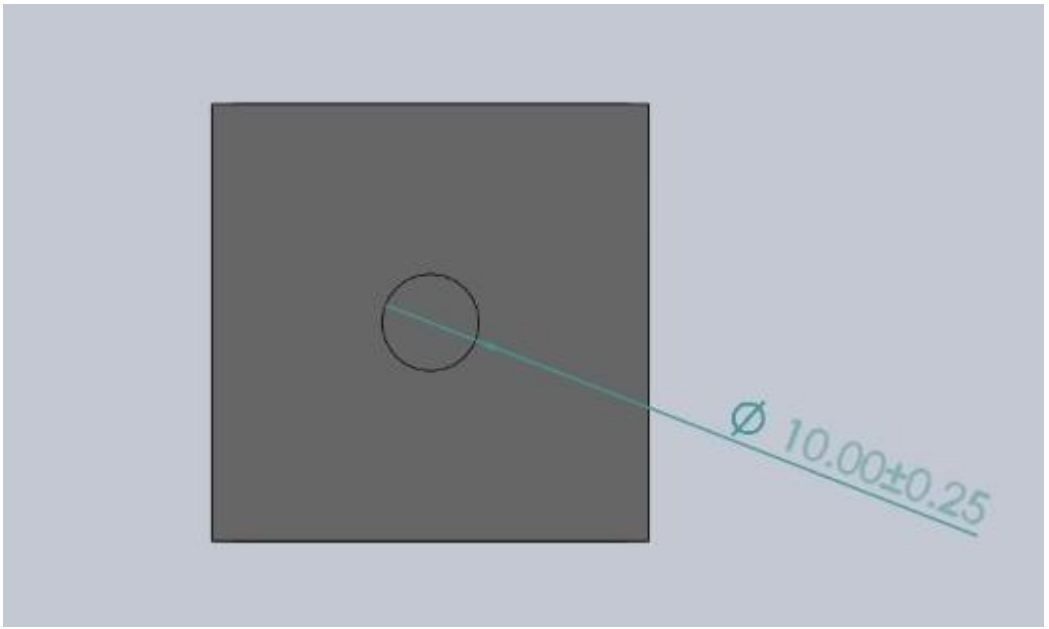


2.	Metal Plate	A. Cutting of metal base plate from metal sheet to required dimensions. B. Grinding the sharp edges after cutting. C. Drilling 3 holes to bolt the wood and metal plate.	A. Plate: 100x100x3 B. Holes: Dia 6(2 holes) and dia 10(1 hole) .	A. Hacksaw cutter B. Grinding Wheel C. Drilling Machine

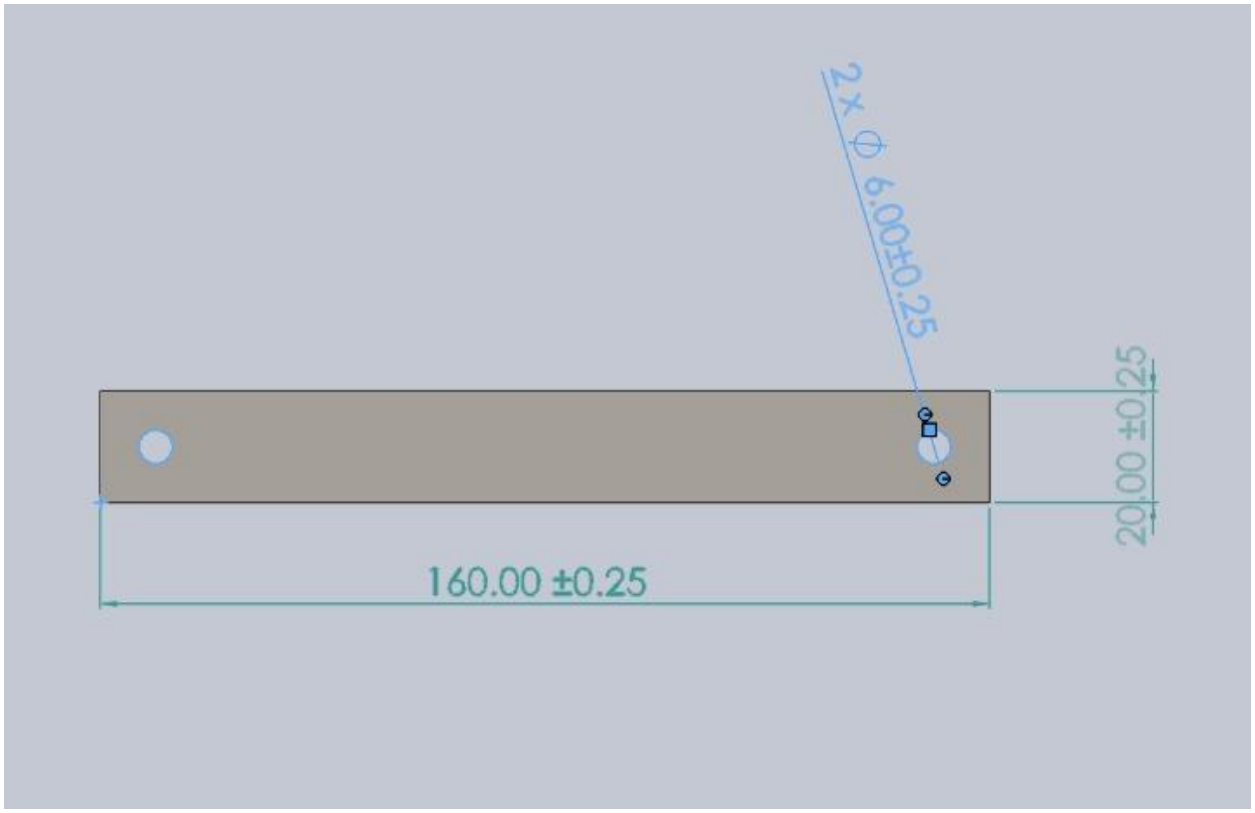


3.	Blocks	A. Take a hollow square cross-section pipe and cut it to dimensions shown. B. Making three bolt positions to bolt with upper, lower arms and base plate.	A. Block; 40x.5x40 B. Holes; Dia 10(1 hole) and Dia 6(2 hole).	A. Hacksaw Cutter B. Drilling Machine

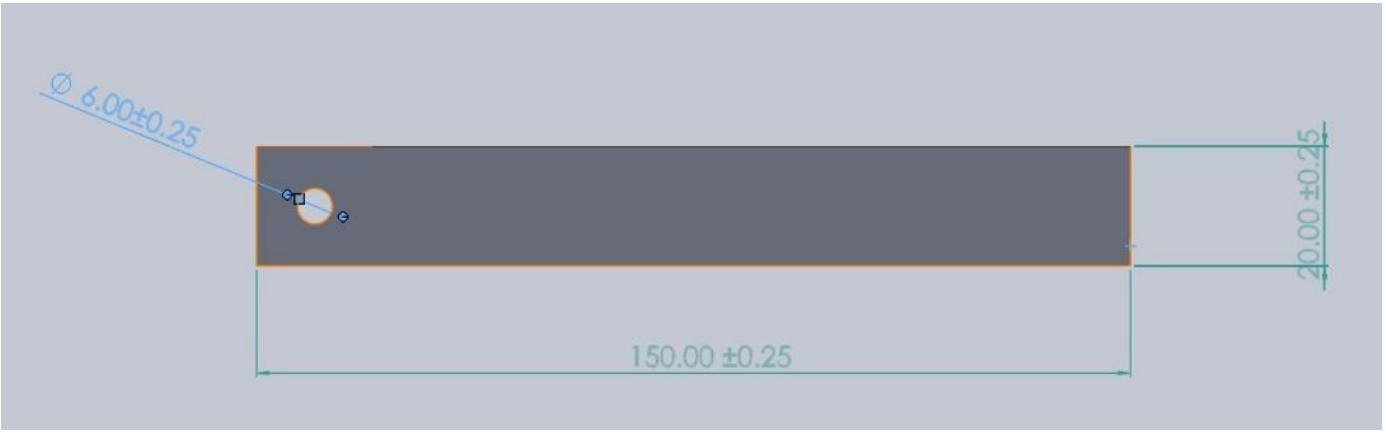




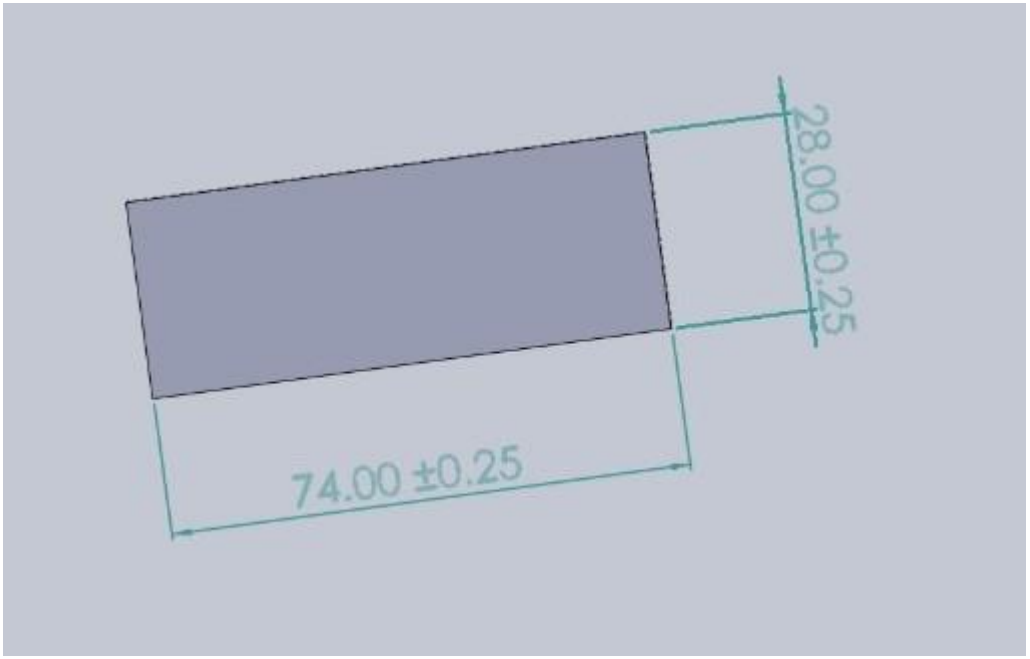
4.	Lower Arm	A. Cut from a metal strip of required breadth. B. Drilled holes on both sides C. Using a grinding wheel, we grind its sharp edges.	A. Arms: 150x20x3 B. Holes: Dia 6(2 holes)	A. Hacksaw Cutting B. Table drill C. Grinding machine



5.	Upper Arm	A. Cut from a metal strip of required breadth. B. Drilled holes on one side. C. Using a grinding wheel, we grind sharp edges.	A. Arms: 150x20x3 B. Holes: Dia 6(1 hole)	A. Hacksaw Cutting B. Drilling Machine C. Grinding machine



5.	Holding Plate	A. Cut a small plate to required dimensions. B. Then we welded it with our upper arms.	A. Plate: 74x28x4	A. Hacksaw Cutting B. Welding machine of 200A range and DC output current



ASSEMBLY

